Overview of Technical Analysis

Two Types of Technical Analysis

- Conventional Statistical Analysis

- Situational Analysis

Conventional Analysis

- It is the statistical analysis you learn at university.
- Multiple regression, logistic regression and neural network all belong to this category.
- It requires a fair amount of past data.
- You build models based on the past data.
- Then you select the best model to predict the future.
- This type of analysis relies on the assumption that the pattern in the future will be quite similar to the pattern in the past.
- This assumption is often violated in the stock market.

Situational Analysis

- It is to take the temperature as it is.
- That is, it tries to understand the current situation of a given stock (or the market) instead of predicting the future.
- We will take the most appropriate action based on our understanding of the current situation.
- It only requires the most recent data.
- It does not rely on the assumption that the future patterns will be similar to past patterns.
- This type of analysis might be more effective than conventional analysis.
- It would be beneficial to use situational analysis in conjunction with conventional analysis.

Daily Investment Exercise

- Your investment decisions need to be made based on the closing prices of stocks on November 6, which is a Friday.

- Return on daily investment will be calculated using the closing prices of the chosen stocks on November 9, which is a Monday.

- Three types of stock price movement.

- Pick two types out of three.

Reversal versus Momentum

How to characterize reversal strength?

Reversal Strategy

- Sequential subtraction for daily change in price?

- Descaling is important!

How to characterize momentum strength?

Momentum Strategy

- Binomial statistic for momentum strength?
- Percentage increase during the observation period? (e.g., past week, or past two weeks)
- If a stock has gone up a lot during the observation period, is it more likely (or less likely) to go up on November 9?
- Is the binomial statistic the higher the better?
- Backtest and forward test are extremely important!
- How to combine the two statistics into one?

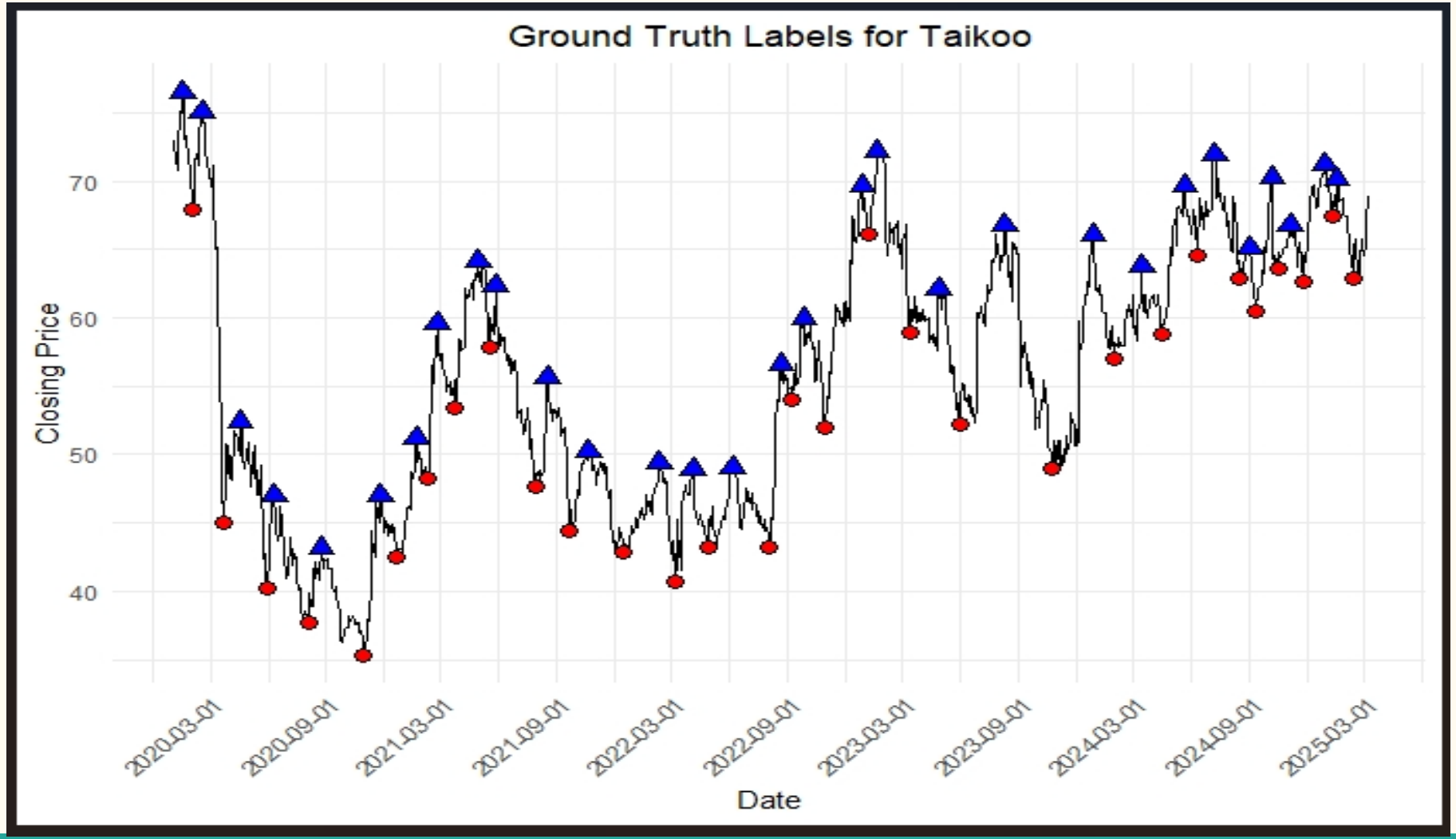
Daily versus Weekly Investment

- The analysis for daily investment can also be applied to weekly investment. And all we need to do is to change the definition of t from a given trading day to a given trading week.

- However, we will lose a lot of granular information if we do that.

- Another approach is to take the price movement cycle into account.

- Again, two approaches (or more approaches) can be used together to have a better understanding of the whole picture.


Ground Truth Labels for Taikoo
Closing Price
70
60
50
40
2020-03-01
2020-09-01
2021-03-01
2021-09-01
2022-03-01
2022-09-01
2023-03-01
2023-09-01
2024-03-01
2024-09-01
2025-03-01
Date

Price Movement Cycle

- How long does it take to complete a full cycle?
- How long does it take to move from the bottom (i.e., trough or valley) to the peak?
- Where are we in the cycle? Are we in the upward movement part of a cycle? Or the downward movement part of a cycle?
- If it usually takes 8 weeks to climb from the bottom to the peak and we have touched the bottom two weeks ago, would it be safe to assume that the price should be higher in a week's time?

Time-series Analysis

- Time-series analysis could be applied to weekly and monthly investments.
- Time-series analysis tries to answer the following two questions.
    - How long does it take to move from the bottom to the peak?
    - What will be the price difference between the peak and the bottom?
- To answer these questions, we could model the following variables:
    - Closing price (less effective)
    - The price difference between adjacent peak and bottom
    - The number of days between adjacent peak and bottom

Time-series Analysis cont.

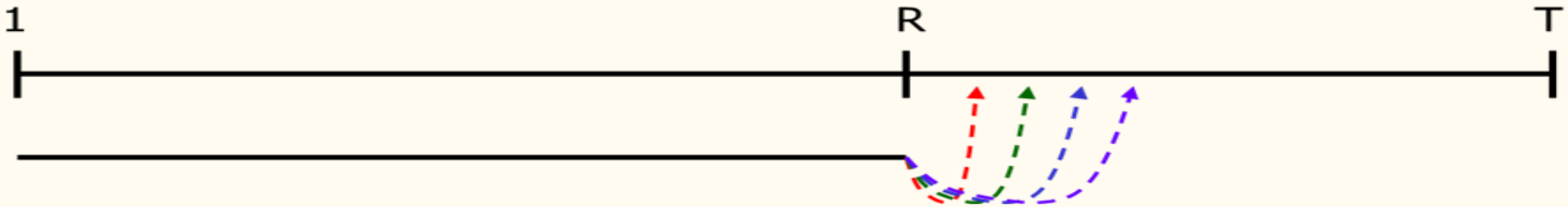
- To do that, we need to generate the ground truth labels for peaks and bottoms.
- We need to have the definitions for training and validation periods.
- The testing period is either between Nov 9 and Nov 13, or between Nov 9 and Dec 4.
- We also need to define the loss metric, e.g., time loss and price loss.
- We also need to select a validation scheme.

Three Types of Validation Schemes

- The Fixed Scheme
- The Recursive Scheme
- The Rolling Scheme

The Fixed Scheme

- Train the models using the data from the training period, say, between January 1st , 2020 and December 31st , 2025.
- For each model that you have trained, produce forecasts for the “peaks” and “bottoms” during the validation period, say, between January 1st , 2026 and October 30th , 2026.
- The fixed scheme is the most demanding forecasting scheme because the model is not updated during the entire validation period.

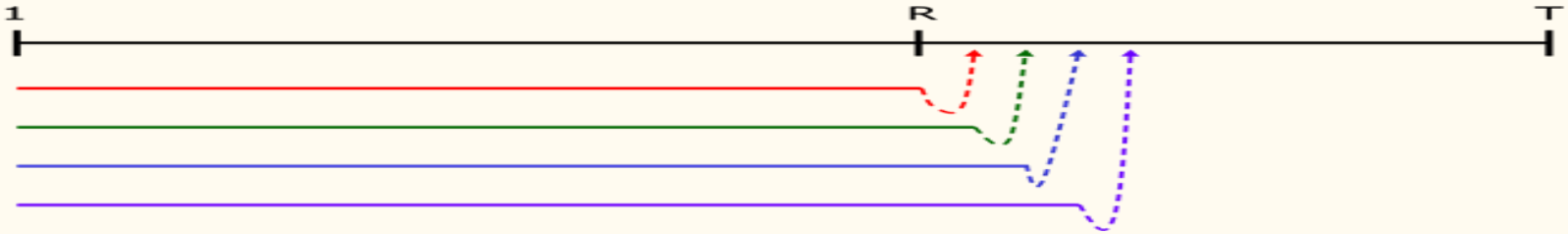


The Recursive Scheme

- Train the first-stage models using the data from the training period, say, between January 1st , 2020 and December 31st , 2025.
- For each model trained, produce a forecast for the next period, i.e., $(R + 1)$th period.
- Include the next period's data into your training dataset, retrain the model and produce another forecast for a future period, i.e., the $(R + 2)$th period.
- Include the $(R + 2)$th period's data into your training dataset, retrain the model and produce a forecast for the future period, i.e., $(R + 3)$th period.
- And so forth...

The Recursive Scheme cont.

- The recursive scheme is less demanding than the fixed scheme due to a shorter forecasting period, and models are retrained and updated during the validation process.



The Rolling Scheme

- Suppose we fix the size of the sample used for forecasting.
- Train the first-stage models using the data from the training period, say, between January 1st , 2020 and December 31st , 2025.
- For each model trained, produce a forecast for the next period, $(R + 1)$th period.
- Include $(R + 1)$th period's data while dropping the first period's data from the training dataset, retrain the model and produce a forecast for the future period, i.e., the $(R + 2)$th period.
- Include $(R + 2)$th period's data while dropping the second period's data from the training dataset, retrain the model and produce a forecast for the future period, i.e., the $(R + 3)$th period.
- And so forth...

The Rolling Scheme cont.

- The rolling scheme is named as such because a fixed-size sample continuously rolls forward to generate forecasts for a fixed horizon.



Fixed, Recursive or Rolling

- Again, the fixed scheme is the most demanding among the three.
- The recursive scheme has a similar intuition to the way we humans produce forecasts.
- Note that in the rolling scheme, the information at the beginning of the last sample is ignored as we roll along.
- Why do we want to throw away some information when such information is in fact available to us?
- One reason is that rolling scheme can help avoid the influence of structural breaks in the data.